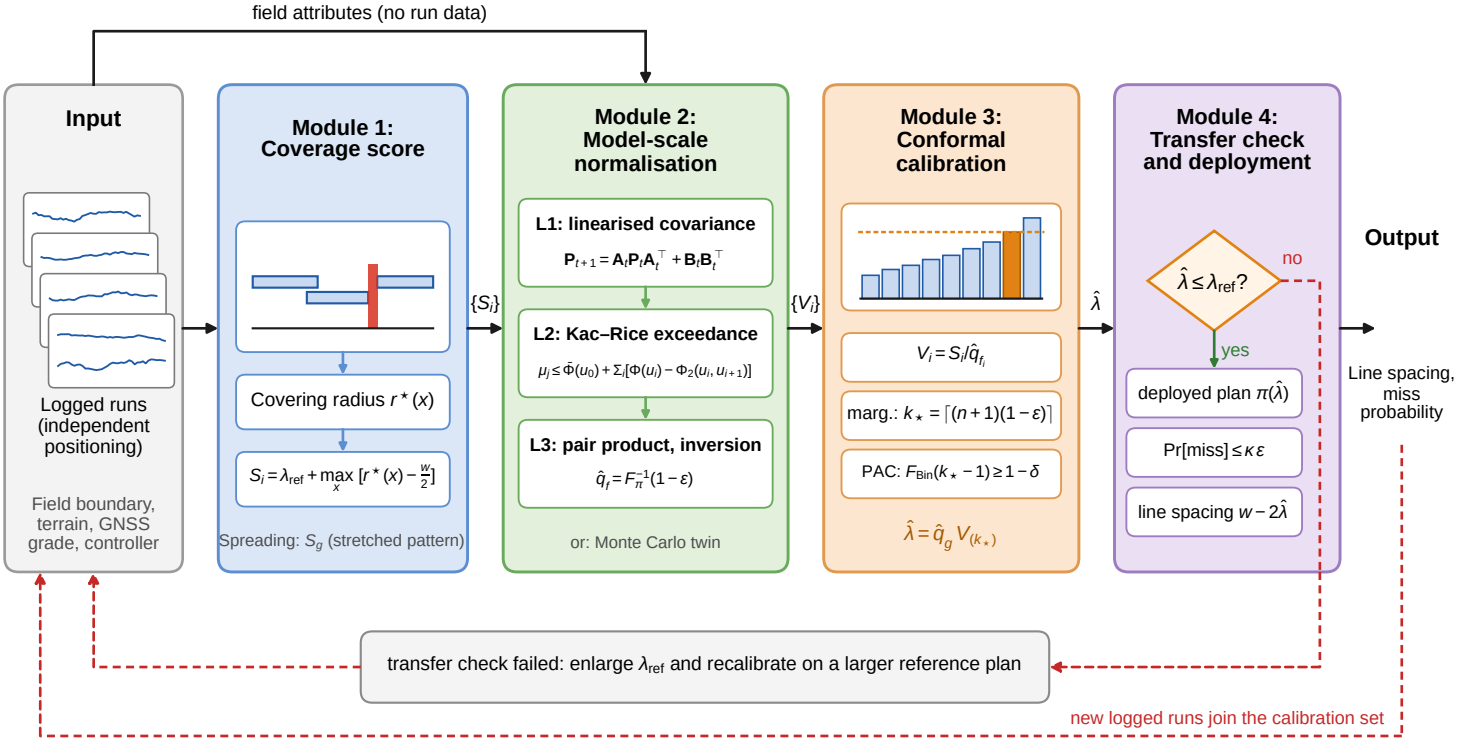
field attributes (no run data)
Input
Logged runs (independent positioning)
Field boundary, terrain, GNSS grade, controller
Module 1: Coverage score
Covering radius $r^\star(x)$
$S_i = \lambda_{\text{ref}} + \max_x [r^\star(x) - \frac{w}{2}]$
Spreading: S_g (stretched pattern)
$\{S_i\}$
Module 2: Model-scale normalisation
L1: linearised covariance
$\mathbf{P}_{t+1} = \mathbf{A}_t \mathbf{P}_t \mathbf{A}_t^\top + \mathbf{B}_t \mathbf{B}_t^\top$
L2: Kac–Rice exceedance
$\mu_j \le \bar{\Phi}(u_0) + \Sigma_i [\Phi(u_i) - \Phi_2(u_i, u_{i+1})]$
L3: pair product, inversion
$\hat{q}_f = F_\pi^{-1}(1-\varepsilon)$
or: Monte Carlo twin
$\{V_i\}$
Module 3: Conformal calibration
$V_i = S_i / \hat{q}_{f_i}$
marg.: $k_\star = \lceil (n+1)(1-\varepsilon) \rceil$
PAC: $F_{\text{Bin}}(k_\star - 1) \ge 1 - \delta$
$\hat{\lambda} = \hat{q}_g V_{(k_\star)}$
$\hat{\lambda}$
Module 4: Transfer check and deployment
$\hat{\lambda} \le \lambda_{\text{ref}}$?
no
yes
deployed plan $\pi(\hat{\lambda})$
Pr[miss] $\le \kappa\varepsilon$
line spacing $w - 2\hat{\lambda}$
Output
Line spacing, miss probability
transfer check failed: enlarge λ_{ref} and recalibrate on a larger reference plan
new logged runs join the calibration set